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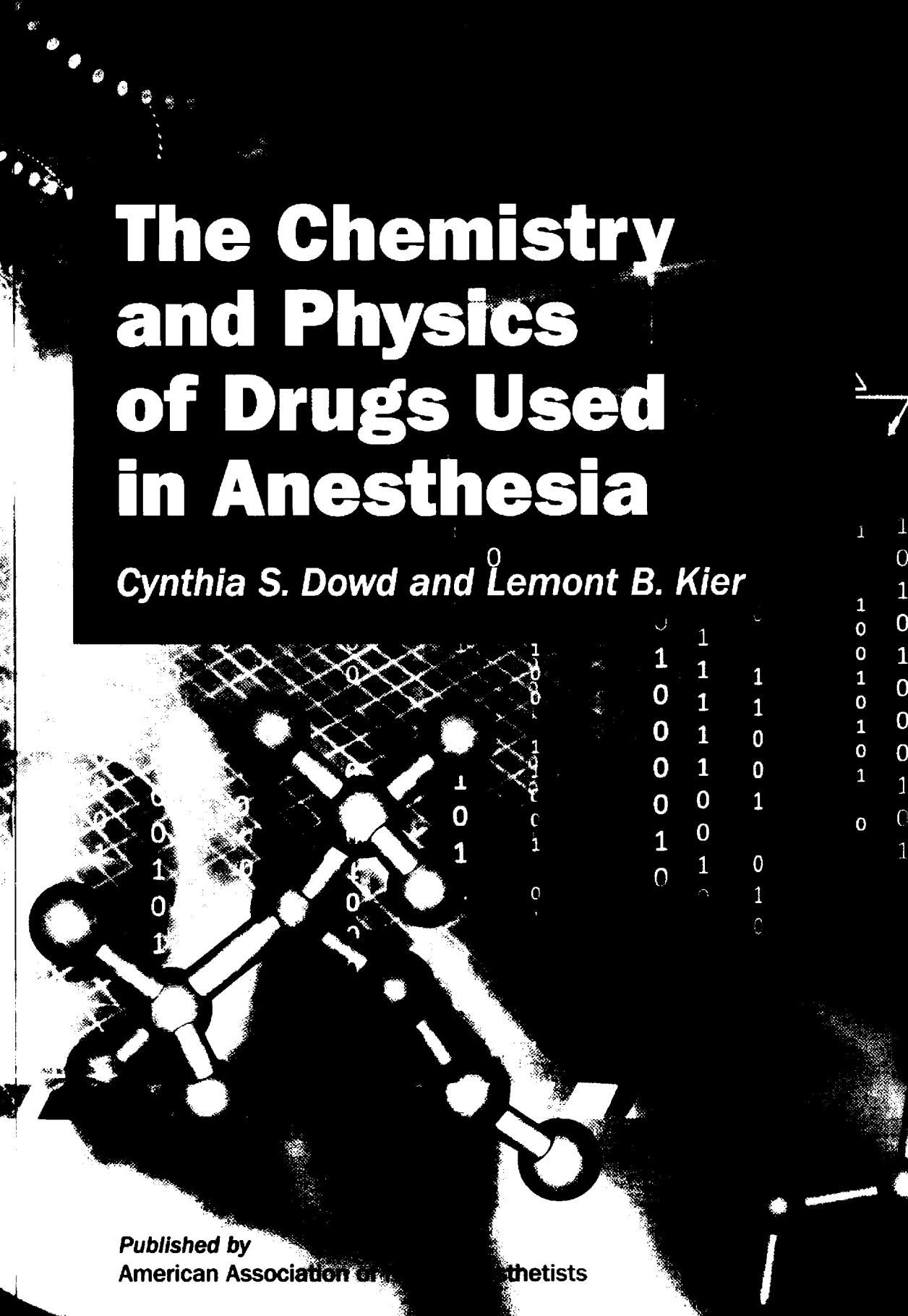
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The Chemistry and Physics of Drugs Used in Anesthesia

Cynthia S. Dowd and Lemont B. Kier



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The Chemistry and Physics of Drugs Used in Anesthesia

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Chapter 8

Cardiovascular Agents

by Umesh R. Desai, PhD, Rami A. Al-Horani, PhD, Akul Y. Mehta, PhD, and May H. Abdel Aziz, PhD

Cardiovascular disease is the most prevalent disease of the developed and developing world and accounts for the largest health burden of numerous countries. It is estimated that more than 50% of deaths are due to some form of cardiovascular disease. A substantial proportion of these deaths can be prevented through lifestyle modifications coupled with appropriate therapeutic treatment. Cardiovascular drugs can be broadly categorized as follows: (1) antianginals, (2) antiarrhythmics, (3) antihypertensives, (4) anticoagulants, (5) antihyperlipidemics, (6) hypoglycemics, and (7) antithyroid drugs and thyroid hormones. This chapter focuses on the first 4 classes of cardiovascular agents.

Antianginal Drugs

Antianginal drugs are used to treat angina pectoris. There are 3 classes of agents that relieve anginal pain: *organic nitrates* and *calcium channel antagonists* are indicated in spasmodic and chronic stable angina, whereas β-*adrenergic antagonists* are primarily used for exertion-induced angina. Antianginal agents mainly alleviate the pain by reducing the oxygen requirements of the heart, thereby reducing anginal pain. Each class of antianginal agent uses a distinct mechanism for reducing the heart workload, and, consequently, 2 or more antianginal agents may be simultaneously used to increase the therapeutic effect.

Organic Nitrates

Organic nitrates (**Figure 1**) are also called *nitrovasodilators.* Amyl nitrite, discovered in 1867, was the first antianginal agent. Several organic nitrates with varying potency are now available for clinical use. Although newer agents such as the calcium channel antagonists and β-adrenergic antagonists have been introduced, organic nitrates remain the drugs of choice for treating spasmodic episodes of angina.

Chemistry

Nitrovasodilators are small nitrate or nitrite esters of simple organic alcohols. Whereas normal organic esters, eg, RCOOR', are a combination of an organic acid (RCOOH) with an organic alcohol (R'OH), the nitrovasodilators are esters of nitrous (HNO_2) or nitric (HNO_3) acid with an organic alcohol (**Figure 2**). It is important to note that all nitrate (and nitrite) esters consist of an O—N bond,

and not a C—N bond. The name *nitroglycerin* (Figure 1) seems to indicate the presence of a nitro group (NO_2) attached to an alkyl carbon, but it is a misnomer and should be more appropriately called glyceryl trinitrate. Also, amyl nitrite consists of an isoamyl group and should be more correctly called isoamyl nitrite.

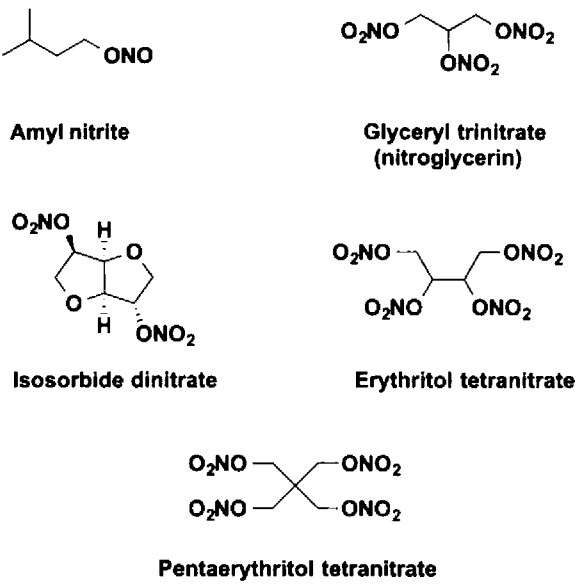


Figure 1. *Organic nitrates are nitrovasodilators and are used as antianginal agents.*

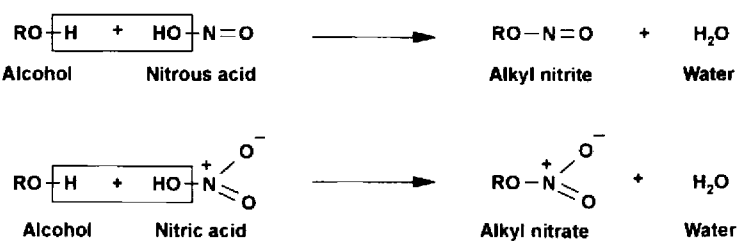


Figure 2. *Formation of organic nitrates by esterification of nitrous (or nitric) acid with an alcohol.*

As shown in structures in Figure 1, the nitrovasodilators are small uncharged organic molecules. A specific advantage results from this characteristic. Because of their nonpolar nature, these agents exhibit high lipid permeability. Thus, rapid treatment of acute anginal episodes is possible through fast absorption. The severe anginal pain is relieved within minutes. Most agents are fairly volatile, causing some concern in handling. Being esters, nitrovasodilators are susceptible to hydrolysis, and, hence, long-term storage is a concern because of the potential for loss of activity. Preparations of these agents should be protected from moisture. In addition, these nitrate esters also exhibit potential for explosion. Thus, many of the agents are available in diluted forms in the presence of excipients that minimize this hazard.

Pharmacokinetics

The onset and duration of action of these agents depend on the structure of the molecules. The smallest agent, amyl nitrite, a gas, can be inhaled and, hence, is the one that shows almost instantaneous effect on administration (30 seconds). Although sublingual and oral modes of administration are available, in general, the larger the molecule and more sterically hindered the nitrate group, the longer the onset and the duration of action. Thus, glyceryl trinitrate and isosorbide dinitrate have a shorter onset time (< 5 minutes) in comparison with erythritol tetranitrate and pentaerythritol tetranitrate (15-30 minutes). Similarly, the duration of action is 30 to 60 minutes for smaller molecules and 3 to 5 hours for the larger molecules.

Metabolism

Organic nitrates are rapidly metabolized by first-pass metabolism in the liver and also extrahepatic tissues such as bloodstream, kidneys, lungs, and intestinal mucosa. The metabolism of the organic nitrates is the principal reason for their action as antianginal agents. In this process, the organic nitrates react with cysteine-containing proteins, resulting in the release of nitric oxide, NO, that is responsible for the vasodilating effect on the arteries. Thus, the parent organic nitrates do not have inherent antianginal activity and can be viewed as *prodrugs,* agents that release the therapeutically active entity in the human body. Chemical and enzymatic processes release NO in situ from the nitrovasodilators. Chemical agents such as cysteine react with organic nitrates to form S-nitrosothiols (R-S-NO) that decompose rapidly to release NO, whereas glutathione-nitrate reductase is a specific enzyme that reduces the organic nitrates to nitrites that subsequently release NO nonenzymatically.

Biochemical Mechanism of Action

Figure 3 depicts the biochemical events that regulate the contraction and relaxation function of all muscle (smooth, cardiac, and skeletal). The state of muscle (contraction or relaxation) is controlled by the action of myosin-actin pairs of proteins. Depending on whether myosin is phosphorylated, the action

of actin results in contraction or relaxation of the muscle. The NO released by nitrovasodilators activates guanylate cyclase, an enzyme that produces cyclic guanosine monophosphate, or cGMP. An increase in the concentration of cGMP, in turn, activates protein kinases that phosphorylate myosin light-chain kinase, thus preventing the phosphorylation of myosin and resulting in muscle relaxation. Muscle relaxation or vasodilation results in reduced workload for the heart, thus easing the anginal pain.

Calcium Channel Antagonists

Cellular levels of free calcium ions have an important role in the mechanism of muscle contraction. Molecules that block the passage of calcium ions from the outside to the inside of the muscle cell (Figure 3) will also prevent the contraction of muscles, leading to reduced workload and, hence, a lower oxygen requirement. Of 6 types of voltage-gated calcium channels, an L-type channel, named for its long-lasting nature, is principally responsible for the inward current of divalent calcium ions into skeletal, cardiac, and smooth muscle cells. Calcium channel antagonists that bind these L-type channels cause antagonism and are effective as antianginal agents. These agents do not physically block the channel, but bind at specific sites in the open form of the channel.

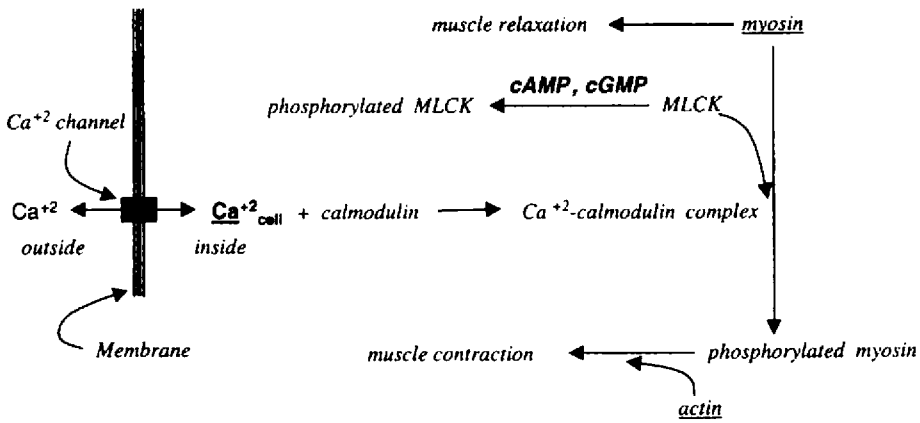


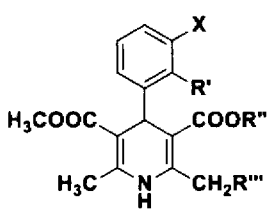
Figure 3. *Regulation of muscle contraction and relaxation. Abbreviations: cAMP, cyclic adenosine monophosphate; cGMP, cyclic guanosine monophosphate; MLCK, myosin light-chain kinase.*

Chemistry

Three classes of calcium channel blockers are currently approved for use in the prophylactic treatment of angina: the dihydropyridines (**Figure 4**); the benzo-1,4-thiazepine, diltiazem (**Figure 5**); and the arylalkylamines (**Figure 6**). No structural similarities exist among the 3 classes of compounds, suggesting that the activity profile of each class is distinct from the others. Nifedipine, amlodipine, and nicardipine belong to the dihydropyridine class of calcium channel blockers (Figure 4). These drugs have a substituted pyridine ring that is partially saturated as a central common feature. Diltiazem belongs to the benzo[*b*-1,5]-thiazepine family, bearing a 7-membered ring containing nitrogen and sulfur atoms fused with an aromatic ring (Figure 5). The arylalkylamines, verapamil and bepridil (Figure 6), have only one thing in common, an amine group substituted with an alkyl and an aryl group. The arylalkylamines have a chiral center, where the dextrorotatory isomer is more active than its counterpart. These drugs are extensively bound to plasma proteins and are metabolized in the liver to form mostly inactive metabolites, which are excreted primarily in urine, except for norverapamil, which retains about 20% of the activity of the parent drug. Norverapamil is formed by removal of the methyl (CH_3) group from the tertiary amine of verapamil.

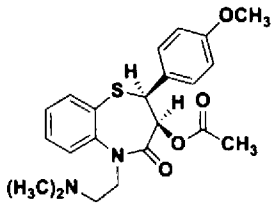
Pharmacokinetics

Each calcium channel blocker contains an amine group facilitating the preparation of its hydrochloride salt for administration as oral tablets and capsules. Also, these agents have a predominantly hydrophobic structure, explaining their rapid and complete absorption after oral administration. In fact, nearly 75% to 95% of the drug is found in the bloodstream. Most of these agents exist primarily in the protein-bound state (80%-95%) in the plasma, although they are active in the free form. The duration of action ranges from 4 to 8 hours for most agents, except for amlodipine, which has a 24-hour duration of action because of the presence of the chlorine atom.



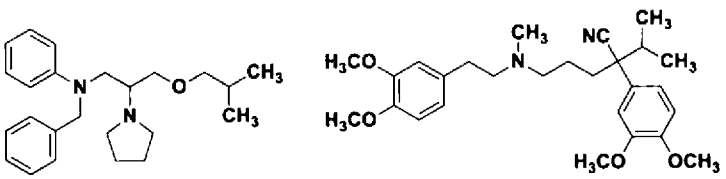
	R'	R"	R'"	X
Nifedipine	$-NO_2$	$-CH_3$	-H	-H
Amlodipine	-Cl	$-C_2H_5$	$-O-(CH_2)_2-NH_2$	-H
Nicardipine	-H	$-(CH_2)_2N(CH_3)(CH_2Ph)$	-H	$-NO_2$

Figure 4. *The dihydropyridines act as calcium channel blockers.*



Diltiazem

Figure 5. *The structure of the calcium channel blocker, diltiazem.*



Bepridil Verapamil

Figure 6. *The arylalkylamines act as calcium channel blockers.*

Metabolism

The first-pass metabolism of verapamil, diltiazem, nicardipine, and nifedipine is extensive and results in low bioavailability. Verapamil is converted into norverapamil in which the nitrogen has been *N*-demethylated. Norverapamil is only about 20% as active as the parent active molecule. Extensive *O*-demethylation also occurs, rapidly giving inactive metabolites. Diltiazem is metabolized by the action of esterases to its desacetyl derivative that has only about 50% of the parent activity. Other *N*- and *O*-demethylations result in inactive metabolites. The dihydropyridines are mostly metabolized to inactive species in which the phenyl group has been extensively hydroxylated.

β-Adrenergic Antagonists

Propranolol (**Figure 7**) is a common nonselective β-blocker of cardiac and bronchial adrenergic receptors. It is typically used for exertion-induced angina that originates from coronary atherosclerosis. Drugs with β-blocking activity slow the heart rate and decrease the force of contraction of muscles; thus, these

drugs are useful in treating hypertension and cardiac arrhythmias, in addition to angina. Propranolol is also typically used in combination with organic nitrates or calcium channel blockers to enhance its antianginal efficacy. Some newer β-blockers, such as carvedilol, also have partial α_1-adrenergic blocking effects that cause vasodilation. Others, like metoprolol, have antiarrhythmic effects similar to those of propranolol.

Newer Nontraditional Anti-ischemic Agents

Ranolazine (**Figure 8**) is the first antianginal drug to be approved by the US Food and Drug Administration in more than 25 years. It exhibits a new mechanism of action by inhibiting the slow sodium channel in ventricular heart muscle, leading to the decrease in the inward sodium current. Sodium-calcium exchange is thus reduced, and calcium accumulation in the heart muscle cells is indirectly prevented. Unlike β-blockers, nitrates, and calcium channel blockers, ranolazine does not affect the heart rate or blood pressure.

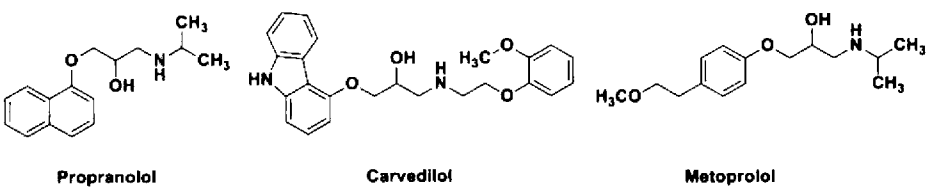


Figure 7. *β-Adrenergic agents used as β-blockers.*

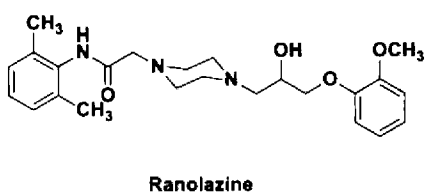


Figure 8. *The structure of ranolazine, an inhibitor of the slow sodium channel in cardiac muscle.*

Antiarrhythmic Agents

Cardiac arrhythmias can originate from a disturbed origin of the impulse, ie, pacemaker cells. Altered automaticity sometimes develops in the pacemaker cells, resulting in disturbance of the rhythmic beating property arising from optimal membrane depolarization. Disturbed automaticity of pacemaker cells may arise from underlying diseases such as hypertension, atherosclerosis, hyperthyroidism, and lung disease. Other forms of arrhythmias, called ectopic arrhythmias, may be caused by generation of impulses in cells other than pacemaker cells. The underlying causes of ectopic arrhythmias are myocardial ischemia, excessive myocardial catecholamine release, and the toxic effects of cardiac glycosides. Arrhythmias are also produced when the electrical impulse does not die down completely before the beginning of phase 0 (**Figure 9**). In such circumstances, a fraction of previous impulse that remains at the end reenters and reexcites the heart muscles, prematurely giving rise to asynchronous depolarization. This is the characteristic form of premature heartbeat. Reentrant arrhythmias are common in coronary atherosclerosis.

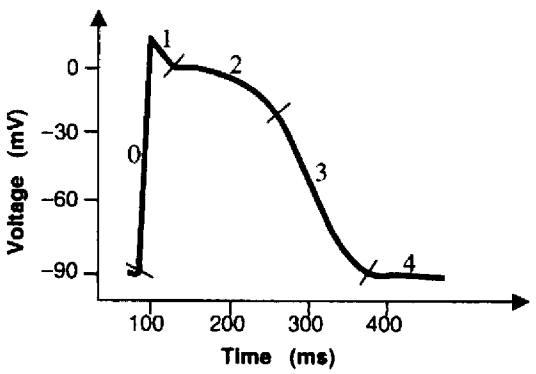


Figure 9. *A representation of the membrane action potential (upper trace) and an electrocardiogram (lower trace). Phase 0 corresponds to rapid depolarization (inward movement of Na^+ ions), whereas phases 1 through 4 are repolarizations through movement of K^+, Ca^{2+}, and Cl^- ions. Repolarization is completed during phase 4, the resting phase. The duration of action potential is the total time for the 0-3 segment.*

Classes of Antiarrhythmic Drugs

The Vaughan Williams classification remains the most widely accepted system to categorize antiarrhythmic drugs. This system classifies the antiarrhythmic agents based on their mode of action and, consequently, on their electrocardiogram effect (**Table**). Drugs with class I properties are generally local anesthetics and block the sodium influx during phase 0 of the action potential. Therefore, these drugs tend to decrease the maximal rate of cardiac depolarization and, accordingly, slow the impulse conduction by their direct effect on the cardiac plasma membrane. All agents in this class bind to the fast sodium channels, which exist in 3 distinct states, referred to as resting, opened, and closed. The affinities of class I agents for these 3 states are different, resulting in differential effects on the duration of action potential. Thus, class I drugs are further subclassified into 3 subclasses (IA, IB, IC) based on their effect on the duration of action potential (QT interval, Figure 9). Class IA drugs increase, class IB drugs decrease, and class IC drugs do not change the duration of action potential. Particularly, class IC agents have slow association-dissociation channel kinetics; thus, they have the most potent sodium channel blocking effects.

Class II drugs are β-adrenergic blocking agents that stabilize the membrane or block adrenergic-enhanced phase 4 depolarization. These agents decrease the neurologically initiated automaticity. Thus, the effects of ectopic pacemaker cells are depressed, resulting in slowing of the heart rate.

Table. *Classes of antiarrhythmic drugs.*

Class	Drug	Mode of Action
IA	Quinidine, procainamide, disopyramide	Decreases MRD; lengthens duration of action potential
IB	Lidocaine, tocainide, mexiletine, phenytoin	Decreases MRD; shortens duration of action potential
IC	Encainide, flecainide, moricizine, propafenone	Decreases MRD; no effect on duration of action potential
II	β-Adrenergic blockers (eg, propranolol)	Suppresses adrenergic-induced automaticity
III	Bretylium tosylate, amiodarone, dronedarone, celivarone, sotalol, ibutilide, dofetilide, azimilide	Many mechanisms; lengthens duration of action potential
IV	Calcium channel blockers (eg, diltiazem, verapamil)	Inhibit slow Ca^{+2} channel; lengthen duration of action potential

Abbreviation: MRD, maximal rate of depolarization.

Class III drugs prolong the duration of action potential without altering the maximal rate of depolarization or the resting potential. Several reports classify the drugs belonging to this class as potassium channel blockers; however, drugs in this class have been found to act through many mechanisms that involve calcium, potassium, and chloride transport. Accordingly, several other reports appropriately classify them as inhibitors of repolarization.

Class IV drugs are calcium channel blockers with antiarrhythmic activity. These agents block the slow movement of calcium ions during phase 2, which leads to lengthening the duration of action potential.

Class I: Local Anesthetics, Sodium Channel Blockers

Class IA drugs include quinidine, procainamide, and disopyramide.

Quinidine (**Figure 10**) is a dextrorotatory diastereoisomer of quinine. Quinidine and quinine are obtained from many species of the *Cinchona* plant. Structurally, quinidine is composed of a quinoline ring and the bicyclic quinuclidine ring system. Thus, quinidine contains 2 basic nitrogen atoms, of which the quinuclidine nitrogen is more basic, with an acidic constant (pKa) of about 11. Quinidine is a prototypic antiarrhythmic drug that reduces the sodium ion current by binding to the open ion channel, resulting in depression of automaticity of ectopic foci. It is used to treat supraventricular and ventricular ectopic arrhythmias, atrial and ventricular tachycardia, atrial flutter, and atrial fibrillation.

Quinidine is available as water-soluble sulfate, gluconate, or polygalacturonate salt. Each has slightly different physical and biological absorption properties. Quinidine sulfate is an oral preparation that also can be used intramuscularly. It is rapidly absorbed from the gastrointestinal (GI) tract, and onset of action begins in about 30 minutes. Quinidine gluconate is soluble in water and is mostly used in emergencies when rapid response may be needed that makes oral administration of quinidine sulfate ineffective. Quinidine polygalacturonate gives more stable and uniform blood levels of quinidine.

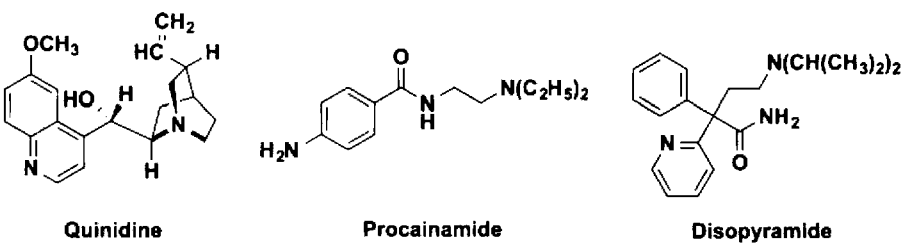


Figure 10. *Class IA antiarrhythmic drugs.*

Procainamide and disopyramide (Figure 10) have emerged as major antiarrhythmic drugs with a mechanism of action similar to that of quinidine. Chemically, procainamide is a *p*-aminobenzamide derivative, whereas disopyramide is a butamide derivative. Both are orally bioavailable, rapidly absorbed from the GI tract, and moderately bound to serum proteins. Metabolism of procainamide results in *N*-acetylprocainamide that has only one-fourth the activity of the parent drug, while half of disopyramide oral dose is excreted in urine unchanged and the other half undergoes *N*-dealkylation through hepatic metabolism. The *N*-dealkylated metabolite of disopyramide retains 50% of its parent antiarrhythmic activity. Most adverse effects of disopyramide are attributed to its antichlolinergic activity.

Some of the class IB drugs are lidocaine, tocainide, mexiletine, and phenytoin. Lidocaine and tocainide are aniline-amide derivatives, whereas mexiletine is an ether derivative (**Figure 11**). Lidocaine has a tertiary amine, and tocainide and mexiletine have a primary amine, which results in a measurable difference in their pKa values (7.7 for tocainide, 9.5 for lidocaine) and affects the degree of the protonated form of these drugs present in the blood. The ether bond of mexiletine imparts greater chemical stability in comparison with tocainide.

Lidocaine was initially introduced as a local anesthetic, but is now routinely used intravenously for treatment of arrhythmias arising from acute myocardial infarction and cardiac surgery. In fact, it is usually the drug of choice for emergency treatment of ventricular arrhythmias. It binds to the active and inactive sodium channels with nearly equivalent affinity, causing depression in diastolic depolarization and automaticity. Lidocaine does not bind to serum proteins to a substantial extent because of its positive charge at the physiological pH. It undergoes extensive first-pass metabolism, which prevents its oral administration. The monoethylglycinexylidide metabolite, resulting from partial de-ethylation of the terminal tertiary amine, is an effective antiarrhythmic agent. Lidocaine has a short half-life of 15 to 30 minutes. Lidocaine solutions containing epinephrine are used only for local anesthetic purposes.

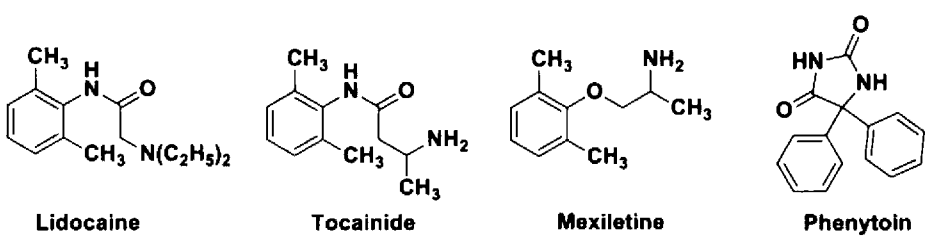


Figure 11. *Class IB antiarrhythmic drugs.*

Tocainide is not subject to first-pass metabolism because of its low hepatic clearance, and, consequently, it is orally bioavailable. It has a plasma half-life of 12 hours, and nearly 50% of the drug is excreted unchanged in urine. It is important to note that it has 2 enantiomers, of which the R isomer is nearly 4 times as potent as the S isomer. It is used orally for the treatment of ventricular ectopy and tachycardia.

Mexiletine has a prolonged half-life on oral administration of nearly 14 hours. Metabolism of mexiletine produces *p*-hydroxy and hydroxymethyl derivatives that are not active as antiarrhythmic agents.

5,5-Diphenylhydantoin (phenytoin, Figure 11) has been used traditionally in the control of generalized tonic clonic seizures (formerly known as grand mal seizures). Phenytoin is structurally analogous to the barbiturates but does not have their sedative effects. Phenytoin is clinically used in treatment of digitalis-induced arrhythmias, and its action is similar to that of lidocaine. Phenytoin may be administered orally; however, it is absorbed slowly. It is extensively plasma protein bound (~90%), and the elimination half-life is between 15 and 30 hours. Parenteral administration is limited to intravenous administration, particularly as a phosphate ester (fosphenytoin).

Class IC antiarrhythmic drugs include encainide, flecainide, moricizine, and propafenone. Encainide and flecainide (**Figure 12**) depress the rate of depolarization and increase the length of refractoriness. Both are benzamide derivatives with local anesthetic properties. Encainide is extensively metabolized into 2 active metabolites. The metabolites 3-methoxy-O^4-demethyl-encainide and O^4-demethyl-encainide are as potent as and more potent than encainide, respectively. Both metabolites can persist in plasma for as long as 12 hours, whereas encainide has a half-life of only 2 to 4 hours. This pharmacokinetic profile leads to a high risk of proarrhythmic adverse effects, which may have been the reason for the withdrawal of this drug from the market. Metabolism of flecainide results in a meta-*O*-dealkylated product with 50% less activity than the parent drug. Flecainide in the acetate form can be given orally to treat ventricular arrhythmias.

Moricizine (Figure 12) is a phenothiazine derivative, oral class IC antiarrhythmic agent that blocks sodium channels. Moricizine has higher affinity for the inactivated state than the activated or resting state of the sodium channels. It is used exclusively for life-threatening ventricular arrhythmias.

Propafenone (Figure 12) can be chemically classified as a phenyl propanone. Although this drug contains a chiral center, it is sold as a racemic mixture. The R and S enantiomers exert a sodium channel blocking effect, but the S isomer also exerts a β-adrenergic effect. The S isomer produces a nearly 40-fold greater antiarrhythmic effect than does the R enantiomer. Metabolism of both enantiomers produces 5-OH metabolites that are as active as the parent compounds. This drug has been primarily proven for ventricular and supraventricular arrhythmias. The drug has poor bioavailability (20%) due to first-pass metabolism and has a short half-life.

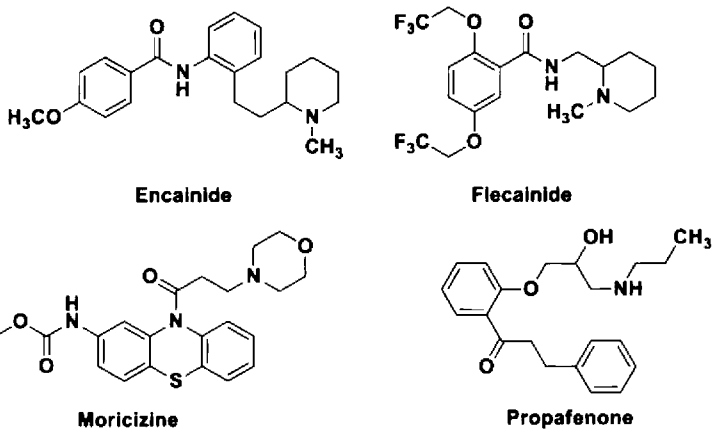


Figure 12. *Class IC antiarrhythmic drugs. Note that encainide has been withdrawn from the market.*

Class II: β-Adrenergic Blockers

Propranolol (Figure 7), a naphthaleneoxy propylamine derivative, is the prototype of class II antiarrhythmics. Both stereoisomers of propranolol are active, with the S isomer the most potent enantiomer. It is a nonselective β-adrenergic blocker and is used for the treatment of supraventricular arrhythmias, including atrial flutter, paroxysmal supraventricular tachycardia, and atrial fibrillation. Propranolol is also effective for the treatment of digitalis-induced ventricular arrhythmias.

Class III: Potassium Channel Blockers, Inhibitors of Repolarization

Bretylium tosylate (**Figure 13**) is a quaternary ammonium salt that is extremely soluble in water and alcohol. Bretylium tosylate is an adrenergic neuronal blocking agent that accumulates in the neurons and displaces norepinephrine. Because of this action, it was used earlier as an antihypertensive agent, but its use was discontinued because of the development of tolerance and pain-related adverse effects. It prolongs the effective refractory period but does not affect the rate of depolarization and, hence, is classified as a class III agent. The precise mechanism of action of bretylium tosylate remains to be elucidated.

Amiodarone and dronedarone (Figure 13) are chemically classified as benzofuran derivatives, and each contains a terminal tertiary amine. An important difference is that amiodarone contains a diiodinated aromatic ring, whereas dronedarone is noniodinated. The iodine atoms introduce thyroid-specific adverse effects for amiodarone. Hypothyroidism occurs in up to 11% of people taking amiodarone. In addition, amiodarone inhibits enzymes of the oxidative system and, hence, interferes with the normal metabolism of

many other drugs. Thus, it is used primarily in hospital settings, especially for life-threatening ventricular arrhythmias. Amiodarone and dronedarone exert effects across all Vaughan Williams classes. Amiodarone lengthens the effective refractory period. The drug has a very slow onset of action (days) and is eliminated very slowly from the body, with a half-life of approximately 1 month following oral doses. The lipophilicity of dronedarone is decreased because of the methane-sulfonamide moiety, resulting in faster elimination in comparison with amiodarone. Dronedarone is effectively absorbed after oral absorption; however, its bioavailability is only about 15% because of the extensive first-pass effect. The oral bioavailability may be increased by concurrent administration of the drug with a high-fat meal.

Chemically, sotalol is a phenylethylamine derivative, whereas ibutilide is a phenyl-butylamine derivative (Figure 13). Both contain a methane-sulfonamide group, and both are sold as racemic mixtures. Only the d(+) enantiomer of sotalol has class III activity, whereas the l(−) isomer has class II (β-adrenergic blockade) and class III (potassium channel blockade) activity. Thus, the 2 isomers slow the heart rate in slightly different ways. Sotalol is used orally as prophylaxis for and treatment of the recurrence of life-threatening ventricular arrhythmias. Sotalol is not bound to plasma proteins, and it is excreted unchanged through the renal excretion mechanism.

Ibutilide lacks β-adrenergic blocking activity in comparison with sotalol, and, consequently, it has less potential to promote adrenergic receptor–mediated adverse effects. Because of its extensive first-pass effect, ibutilide must be given as an intravenous infusion as the fumarate salt. It has been approved by the US Food and Drug Administration for the conversion of recent-onset atrial fibrillation and atrial flutter.

Dofetilide (Figure 13) is a bis-methane-sulfonamide derivative. In contrast with sotalol, it exhibits only class III effects without any β-blocking activity. It is more potent and selective than other drugs in class III. It is used orally to suppress atrial flutter and atrial fibrillation. It is well absorbed from the GI tract with almost 100% bioavailability. It binds plasma proteins moderately, undergoes *N*-dealkylation and *N*-oxidation hepatic metabolism to inactive metabolites, and is mainly excreted unchanged in the urine.

Structurally, azimilide (Figure 13) is unrelated to any drugs in this class and yet shows class III antiarrhythmic effect by blocking potassium channels. It is an imidazolidin-dione with a chlorophenylfuranyl side chain. A phase 3 study of azimilide was recruiting participants as of January 2013 (ClinicalTrials.gov identifier NCT01464476); the estimated study completion date is August 2014.

Class IV: Calcium Channel Blockers (Nondihydropyridines)

Diltiazem and verapamil (Figures 5 and 6, respectively) were discussed earlier in this chapter as calcium channel blockers. In addition to their antianginal effects, these drugs also have antiarrhythmic properties. Pharmacologically, both drugs block the slow inward movement of calcium ions, resulting in relaxation

of arterial smooth muscle and vasodilation and decreased contractility in cardiac myocytes. These agents also slow the atrioventricular node conduction and increase its refractoriness. Clinically, supraventricular tachycardia is the major arrhythmic indication for diltiazem and verapamil. These drugs can also reduce the ventricular rate in atrial fibrillation and atrial flutter.

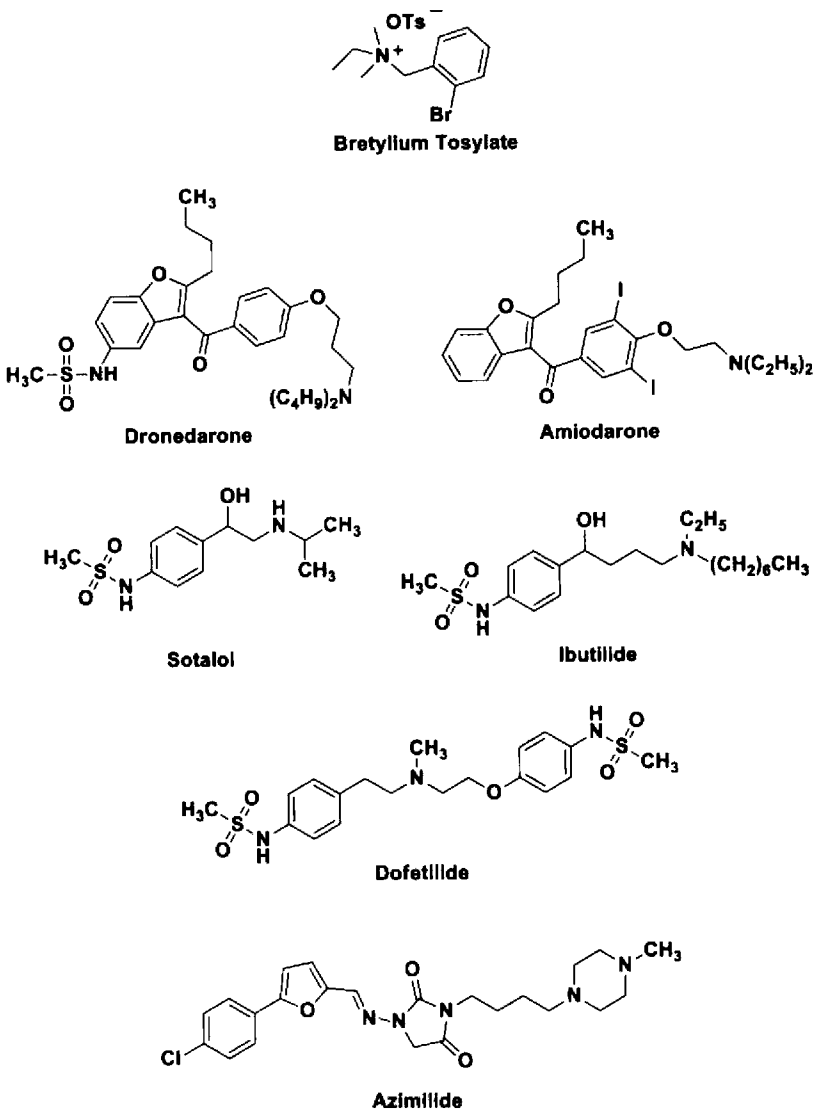


Figure 13. *Class III antiarrhythmic drugs. Note that a phase 3 study of azimilide was recruiting participants as of January 2013 (ClinicalTrials.gov identifier NCT01464476); the estimated study completion date is August 2014.*

Antihypertensive Agents

Hypertension is a known major risk factor for many cardiovascular diseases such as stroke and myocardial infarction. It has been estimated that globally more than 1 billion people are affected by hypertension, with nearly 40 million people affected in the United States alone. Consistently high blood pressure can damage the brain, eyes, and kidneys.

Hypertension is usually classified into 2 types: primary (or essential) and secondary (nonessential). *Primary hypertension,* due to a nonidentifiable cause, is the most common type and occurs in 95% of people who have hypertension. Genetic factors seem to have a role in this hypertension and include genes that regulate a system known collectively as the renin-angiotensin system (RAS) and genes that regulate the sympathetic nervous system.

Secondary hypertension has known causes and can be treated independently and even reversed. This type of hypertension is generally seen during temporary conditions, such as pregnancy; cirrhosis; kidney diseases; and Cushing disease. Other causes include certain medications; lifestyle choices, such as drinking alcohol or excessive drinking of caffeinated drinks such as coffee; and obesity. Temporary hypertension also results from physical stress and exercise.

Targets for Drug Therapies

Hypertension can be combated in a number of ways. Some of the targets include the following:

1. The RAS is responsible for the production of angiotensin II and angiotensin III, which in turn constrict arterioles, causing a rise in the blood pressure. Inhibitors of the RAS are useful antihypertensive drugs.
2. The sympathetic nervous system is responsible for the release of adrenaline and epinephrine, which are neurotransmitters that constrict blood vessels, resulting in increased peripheral vascular resistance. Blocking the effect of these neurotransmitters on the β-receptors present on the blood vessels through the use of β-blockers is a useful strategy for treating hypertension.
3. Voltage-gated calcium channels are responsible for the muscle contractions in the heart and blood vessels. The entry of calcium induces contraction of the blood vessels and heart, thereby increasing the pressure, which can be effectively prevented by calcium channel blockers.
4. Salt reabsorption in the kidneys is responsible for the increase in the volume of fluids in the blood vessels, thereby enhancing blood pressure. This increase in the volume is due to the simultaneous reabsorption of fluids from the kidneys along with salt to maintain osmotic homeostasis. Salt reabsorption can be prevented by the use of diuretic drugs, which causes the body to excrete water and salt.

A discussion of β-blockers and calcium channel blockers was presented earlier in the chapter. Inhibitors of the RAS, diuretics, and direct-acting vasodilators are described in the following paragraphs.

The Renin-Angiotensin System

The RAS (**Figure 14**) is a cascade of enzymatic reactions, which lead to the release of the peptide hormones, angiotensin I and angiotensin II. These peptides are capable of acting on the angiotensin receptors present in the blood vessels and cause vasoconstriction. The angiotensin secreted is also capable of stimulating the adrenal cortex to release aldosterone, a mineralocorticoid hormone responsible for the increased reabsorption of sodium and water.

Reduction of the blood pressure or lowering of the sodium excretion causes a release of the enzyme renin from the kidneys. The enzyme renin then acts on angiotensinogen (released from the liver) to produce angiotensin I, which is activated by angiotensin-converting enzyme (ACE) to produce angiotensin II. Angiotensin II binds to angiotensin receptors, resulting in an increase in the blood pressure. Angiotensin II mainly acts on the arterioles, causing vasoconstriction. It also acts on the kidneys, causing a constriction of the glomerular arterioles, and reduces the glomerular filtration rate, which increases fluid retention. The hormone also stimulates the Na^+/H^+ exchangers located in the renal tubules, causing increased reabsorption of sodium. Finally, it stimulates the adrenal cortex to release more aldosterone, and vasopressin, the antidiuretic hormone, both of which increase the blood pressure. Thus, inhibiting the RAS is important. Three targets have been used for antihypertensive effects: Renin, ACE and the angiotensin receptor. Of these, ACE inhibitors have been used most frequently to reduce hypertension.

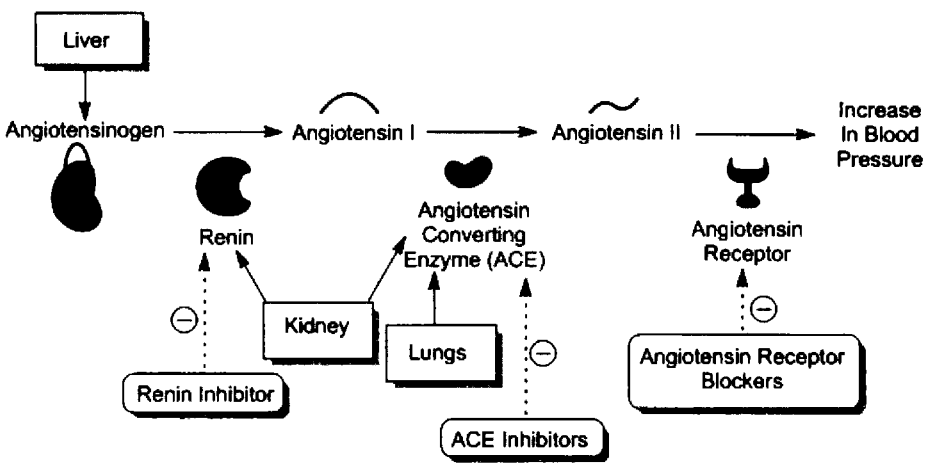


Figure 14. *The renin-angiotensin system (RAS) of blood pressure control. Angiotensinogen is released by the liver. With a rise in blood pressure, renin is released by the kidney, followed by a release of angiotensin-converting enzyme (ACE) by the kidney and lungs. Renin inhibitors block the RAS early in the pathway, and ACE inhibitors inactivate the enzyme and prevent the formation of angiotensin II, which is the key peptide of the RAS. Angiotensin receptor blockers block the activity of angiotensin II at the sites where the peptide shows its effect and thereby block the increase in blood pressure.*

Angiotensin-Converting Enzyme Inhibitors

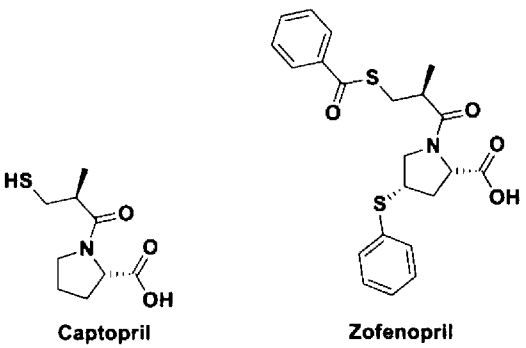
This class of antihypertensives works by inhibiting ACE and preventing the formation of angiotensin II. The ACE inhibitors have names ending with "pril" and are structurally classified into 3 classes.

Class I: Thiol- and Carboxylate-Containing Drugs

Captopril and zofenopril (**Figure 15**) contain a carboxylic acid functional group in their structure, which is recognized by the cationic site arginine in the active site of ACE, but it is not approved. The thiol group was introduced in captopril to increase the binding to zinc of ACE, but it causes side effects such as rash and loss of taste. Zofenopril, on the other hand, has protected thiol groups, which reduces the side effects. Captopril reduces hypertension and is useful in myocardial infarction, congestive heart failure, and even preservation of renal function in diabetic nephropathy. Captopril has a relatively short half-life and poor pharmacokinetic properties. With many adverse effects and poor pharmacokinetics, captopril has reduced patient compliance. Zofenopril has been found to be more effective than enalapril and even atenolol for reducing hypertension. The adverse effects observed in the use of zofenopril are also less severe than with other ACE inhibitors. Zofenopril has been used in other countries for its cardioprotective properties, but it is not approved for use in the United States.

Class II: Aminophenyl Butyric Acid- and Ester-Containing Drugs

This group of drugs contains the 2-(S)-aminophenylbutyric acid functional group (**Figure 16**). These drugs contain 2 carboxylate functional groups, of which 1 interacts with the cationic arginine in the active site. All ester forms of the butyric acid are converted to the active acid form following absorption and metabolism by the liver, kidney, and intestinal enzymes. Hence, it is important for people



Captopril Zofenopril

Figure 15. *Class I angiotensin-converting enzyme inhibitors. Note that zofenopril is not approved for use in the United States.*

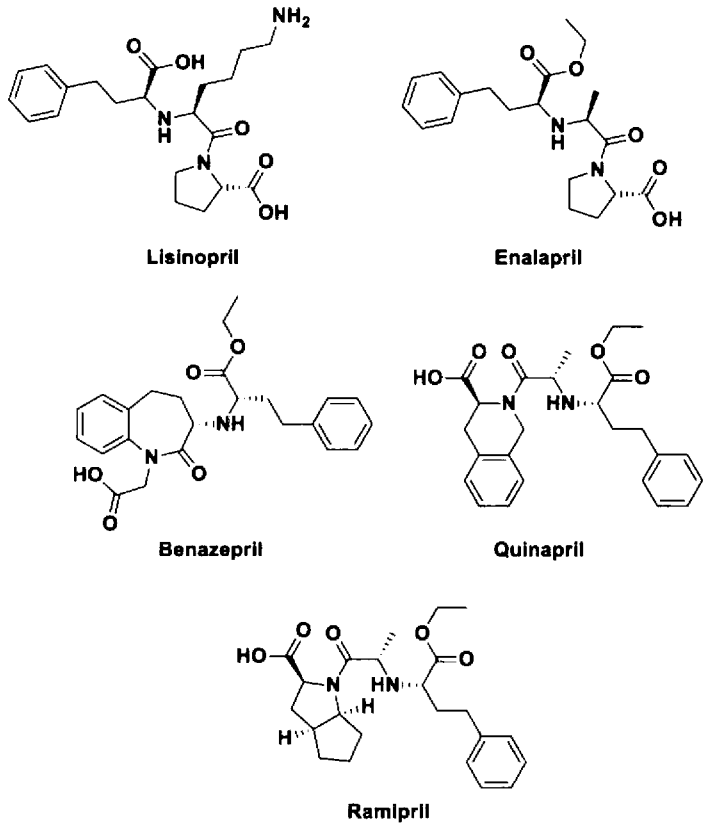


Figure 16. *Class II angiotensin-converting enzyme inhibitors.*

who take these drugs to have active enzymes in these organs; these drugs are contraindicated generally in people with impaired renal and liver function. The ester form of the drug is particularly long lasting because it must be hydrolyzed by nonspecific esterases. These drugs are generally used in mild to moderate hypertension states.

Lisinopril (Figure 16) is so named because it contains a lysine group in its structure, benazepril because of its benzazepine ring, and quinapril on account of its isoquinoline structure. Except for lisinopril, all other agents are prodrugs because their butyric acid is protected as an ester moiety. The cleavage of the ester group produces the active form. Enalapril was the first dicarboxylate containing an ACE inhibitor, but its poor bioavailability led to its conversion to the prodrug form.

Lisinopril is primarily used in congestive heart failure and myocardial infarction and is also useful in preventing renal and retinal complications in diabetes. Dosing is dependent on the creatinine clearance. Enalapril has been

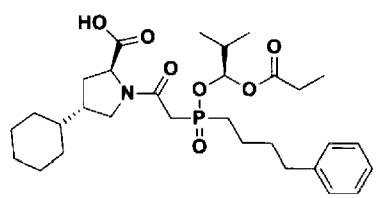
shown to substantially lower the death rates in systolic heart failure. Renal impairment, however, results in the substantial accumulation of enalapril, and, therefore, in such conditions, the dosage must be reduced. Benazepril has been shown to have kidney-protective properties and slows the progression of kidney diseases. Quinapril has been known to cause angioedema and is therefore not widely used. Ramipril is useful mainly as a preventive drug for myocardial infarction and stroke and also to reduce the risk of cardiovascular events during surgical procedures.

Class III: Phosphonate-Containing Drugs

Fosinopril (**Figure 17**) is the only phosphonate-containing inhibitor of ACE. Because of its poor oral bioavailability, the phosphonate group in fosinopril had to be protected with a lipophilic side chain.

Adverse Effects of Angiotensin-Converting Enzyme Inhibitors

The most common adverse effect of ACE inhibitors is persistent dry cough. This cough is believed to be due to an increased bradykinin level, which can also occasionally lead to angioedema. The ACE is also responsible for the breakdown of bradykinin, and, thus, inhibiting it causes an increase in the bradykinin level in the body. If the adverse effect is severe, alternative treatment with angiotensin receptor blockers (see below) is sought. In some cases, renal impairment is also seen, although the reason remains unknown. The ACE inhibitors can also cause hyperkalemia due to a decreased aldosterone level. Potassium supplements are advisable.



Fosinopril

Figure 17. *A class III angiotensin-converting enzyme inhibitor, fosinopril.*

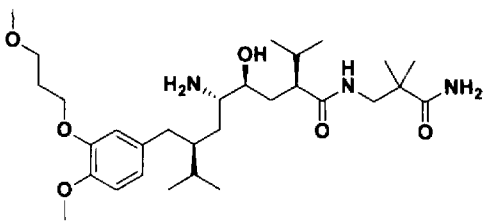
Renin Inhibitors

Renin is an aspartic protease, a rate-limiting enzyme of the RAS. It cleaves angiotensinogen to angiotensin I, a primary catalytic step, which makes inhibition of renin a major strategy in the reduction of hypertension. Other drugs that target the RAS also induce feedback mechanisms to increase renin production, which compensate for the inhibition. However, inhibiting renin directly is devoid of such feedback. The discovery of this enzyme and its role had been known for many years; however, no drugs were designed until recently.

Aliskiren (**Figure 18**), a peptidomimetic drug, binds to the S3bp binding pocket of renin, which is near the active site. Aliskiren is also observed to have renal-protective effects, which are independent of its antihypertensive effects in people with diabetes. On the downside, aliskiren has been known to cause hyperkalemia, particularly when used with ACE inhibitors; angioedema; and hypotension. Other adverse effects observed with aliskiren include gout and kidney stones. Coughing is also observed, although it is not as severe as compared with coughing during use of ACE inhibitors.

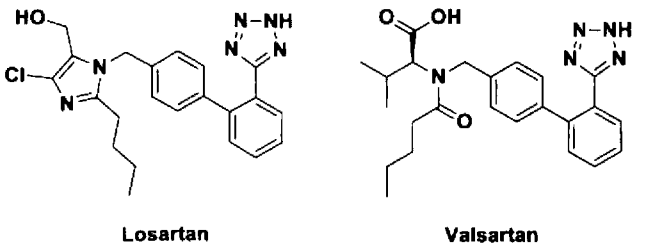
Angiotensin II Receptor Antagonists

This class of antihypertensives works by inhibiting the angiotensin receptors present on the blood vessels. They are also known as angiotensin receptor antagonists, or "sartans" (**Figure 19**). They are mainly useful in hypertension, diabetic nephropathy, and congestive heart failure. Sartans are particularly useful in people who cannot tolerate ACE inhibitors. Valsartan has been one of the most prescribed angiotensin receptor antagonists and has also been indicated for post–myocardial infarction treatment. It is being currently investigated for the treatment and prevention of Alzheimer disease.



Aliskiren

Figure 18. *The structure of aliskiren, a renin inhibitor.*



Losartan Valsartan

Figure 19. *Angiotensin II receptor antagonists.*

Structurally, these drugs generally contain a tetrazole group, and losartan also contains an imidazole ring. The biphenyl linker in both helps binding to the receptor. Losartan, a competitive agonist at the angiotensin II type I receptor, has been seen to decrease the total peripheral resistance and cardiac venous return. Because losartan blocks angiotensin receptors, plasma renin levels increase because of the removal of the negative feedback of the angiotensin II. Mostly, these drugs are well tolerated. Common adverse effects include hyperkalemia due to reduced aldosterone release; angioedema in rare cases; muscle cramps; back pain; insomnia; decreased hemoglobin levels; renal impairment; and nasal congestion.

Diuretics

Kidneys are the main organ of excretion of fluids, small molecules, and ions to maintain homeostasis. If the volume of fluids increases in the blood, the blood pressure increases. Hence, eliminating this excess volume helps in reduction of blood pressure. About one-fifth of the total blood volume is filtered into the nephrons of the kidneys; however, almost 99% of this volume of fluid is reabsorbed and only a small volume is excreted as urine. The amount of water reabsorbed is related, in turn, to the amount of ions that are reabsorbed as the body absorbs the appropriate amount of water to maintain homeostasis. Diuretics can be classified into a number of subtypes based on their targets of action or their structures. Each class of diuretics acts on different parts of the nephron.

Class I: Carbonic Anhydrase Inhibitors

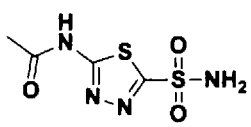
These drugs inhibit the action of carbonic anhydrase, an enzyme found in the proximal convoluted tubule of the nephron. Carbonic anhydrase is an enzyme that catalyzes the reversible conversion of bicarbonate to carbon dioxide and water. The bicarbonate that is excreted recombines with the hydrogen ions in the kidneys to be converted back to carbon dioxide and water, which are reabsorbed by the renal tubule. By inhibiting the carbonic anhydrase, the bicarbonate is not reabsorbed into the blood, which increases urine bicarbonate concentration,

leading to diuresis. Although useful for diuresis, carbonic anhydrase inhibitors are used primarily for the treatment of glaucoma and sometimes for epilepsy. The diuretic action caused by carbonic anhydrase inhibitors is generally compensated by the distal segments of the nephron and, hence, is not very effective.

Acetazolamide (**Figure 20**) is the prototype drug for the carbonic anhydrase inhibitors. Structurally, the drug has a thiadiazole ring along with a sulfamoyl group, which sits in the active site of the carbonic anhydrase. Acetazolamide exerts diuretic action slowly, and the dose must not be increased if there is no loss of edema fluid. The most common adverse effect of acetazolamide is numbing and tingling in the fingers and toes. There can be taste alteration, especially for carbonated drinks. Blurring of vision is also observed in some people, possibly due to reduced intraocular pressure. These drugs can cause metabolic acidosis because they reduce the reabsorption of bicarbonate ions, which might precipitate kidney stones. Dehydration caused by these diuretics could result in headaches and electrolyte imbalance, so there is a need to drink sufficient fluids.

Class II: Loop Diuretics

The loop of Henle reabsorbs a large amount of sodium from the filtrate via the Na^+/K^+-ATPase transporter, which is present on the anti-luminal membrane. This transporter moves 3 Na^+ ions out of the cell of the nephron to the interstitium and 2 K^+ ions in the opposite direction. As a result, there is a deficiency of sodium in the cell, which is compensated by another $Na^+/K^+/Cl^-$ cotransporter channel present on the luminal membrane of the cell cotransporter, which transports 1 Na^+, 1 K^+, and 2 Cl^- ions from the lumen into the cells of the nephron. The return of the K^+ and Cl^- ions down the concentration gradient is achieved via their respective channels. As a result, there is continuous movement of sodium ions from the lumen into the interstitium, causing sodium reabsorption into the body and out of the urine. Loop diuretics act by blocking the $Na^+/K^+/Cl^-$ cotransporter. Even though the sodium-potassium exchange begins with the Na^+/K^+-ATPase transporter, the sodium deficit cannot be compensated by the $Na^+/K^+/Cl^-$ cotransporter, which implies that the overall reabsorption of sodium will be reduced. Enhanced excretion of sodium results in diuresis.



Acetazolamide

Figure 20. *The structure of acetazolamide, a class I diuretic, inhibiting carbonic anhydrase.*

Furosemide (**Figure 21**) is a phenyl sulfonamide–type of loop diuretic, which is primarily used for the treatment of hypertension and the edema associated with renal impairment, heart failure, and hepatic cirrhosis. It is also useful in the treatment of hypercalcemia because it causes excretion of calcium ions as well. There have been some reports of ototoxicity when using furosemide.

Unlike furosemide, ethacrynic acid (Figure 21) is not a sulfonamide, and, thus, it can be used by people with sulfa allergies. Structurally, it is a phenoxyacetic acid derivative that, in addition, contains a ketone and a methylene group. On entering the body, a cysteine adduct of the drug is formed, which, in turn, is the active form of ethacrynic acid. Hence, it may be considered a prodrug. Ethacrynic acid can cause hypokalemia, ototoxicity, and liver damage.

Class III: Thiazide Diuretics

Thiazide diuretics are a class of diuretics that prevent the reabsorption of Na^+ in the distal convoluted tubule. Unlike the loop of Henle, the distal convoluted tubule has a Na^+/Cl^- cotransporter to replace the Na^+ deficit caused by the Na^+/K^+-ATPase transporter. The potassium and chloride ion concentrations are restored by their respective channels. Thiazide and thiazidelike diuretics block the Na^+/Cl^- cotransporter, resulting in the net excretion of the Na^+. While the thiazide diuretics have a benzothiadiazine molecular structure, the thiazidelike diuretics lack this structure but show similar physiological effects.

Chlorothiazide (**Figure 22**) contains a benzothiadiazine ring with a sulfonamide group. It is generally useful in the hospital setting to manage excessive fluids in congestive heart failure, peripheral edema, and hypertension. The presence of a sulfonamide group is a contraindication for the use of chlorothiazide in people with sulfa allergies. Adverse effects often include hypokalemia and dehydration.

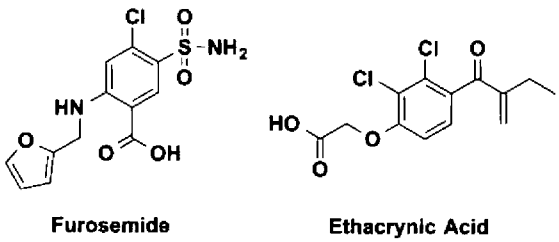


Figure 21. *Class II: loop diuretics.*

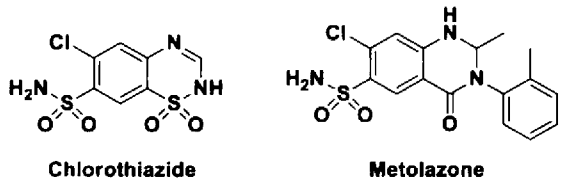


Figure 22. *Class III: thiazide diuretics.*

Metolazone (Figure 22) contains a tetrahydroquinazoline ring, in place of the benzothiadiazine ring, and, hence, it is a thiazidelike diuretic. It is also a sulfonamide and can give rise to allergic reactions. It is used primarily for treating edema associated with congestive heart failure, but is also useful in chronic kidney failure and nephrotic syndrome and is very potent in reducing hypertension. Metolazone can result in hypokalemia, dehydration, aplastic anemia, angioedema, agranulocytosis, and pancreatitis.

Class IV: Potassium-Sparing Diuretics

Potassium-sparing diuretics are a class of drugs that produce diuresis without the secretion of K^+ into the urine. Epithelial sodium channels (ENaCs) are sodium ion channels present in the distal convoluted tubule and the collecting tubules of the nephron. Here, reabsorption of sodium occurs without an anion, thus creating a lumen-negative electrical gradient. This gradient favors the secretion of potassium and hydrogen ions. Inhibition of Na^+ reabsorption by blocking the ENaCs leads to diuresis without the concomitant loss of potassium. These diuretics have 2 basic mechanisms of action: (1) direct inhibition of ENaCs by amiloride and triamterene and (2) aldosterone antagonism, such as by spironolactone and eplerenone.

Amiloride and triamterene (**Figure 23**) are positively charged drugs and can enter and inhibit the ENaCs directly. These drugs can cause kidney stones and may lead to depletion of sodium, folic acid, and calcium. Serious adverse effects include heart palpitations, tingling and numbness, sore throat, rashes, and back pain. The possibility of hyperkalemia also exists. Amiloride is the best tolerated drug in this class of diuretics.

Spironolactone and eplerenone (**Figure 24**) are steroid derivatives. They inhibit the binding of aldosterone to the ENaCs and the activation of the ENaCs by aldosterone. The onset of action of spironolactone may take a few days. Eplerenone is more selective compared with spironolactone and, hence, has fewer adverse effects. As is common with this class of drugs, the possible adverse effects include hyperkalemia, renal impairment, hepatic impairment, and altered renal function.

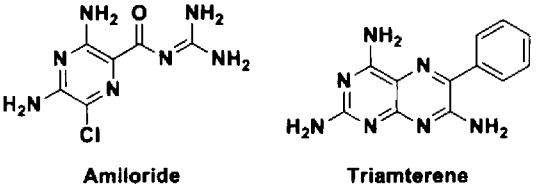


Figure 23. *The structure of amiloride and triamterene.*

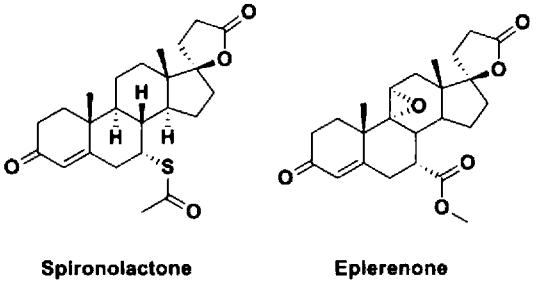


Figure 24. *The structure of spironolactone and eplerenone.*

Class V: Osmotic Diuretics

Certain compounds, such as mannitol (**Figure 25**), are filtered by the nephron, but mannitol is not reabsorbed. This lack of reabsorption increases the osmolality of the filtrate, causing more water to be excreted by the kidneys and leading to diuresis. The use of this class of diuretics is generally not favored.

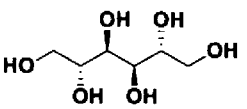
Direct-Acting Vasodilatory Drugs

Drugs that directly induce dilation of the smooth muscle cells (SMCs) are useful in treating hypertension. These drugs include hydralazine, sodium nitroprusside, calcium channel blockers, and potassium channel agonists.

Hydralazine (**Figure 26**) is useful in the treatment of moderate to severe hypertension and is used in combination with other antihypertensive drugs. Hydralazine decreases the action of secondary messenger inositol triphosphate, thereby increasing the amount of guanosine monophosphate levels in the SMCs. This action, in turn, causes limited calcium release from the sarcoplasmic reticulum, causing relaxation and dilation of the SMCs, which reduces resistance to blood flow and decreases blood pressure. In addition, it improves renal blood flow, which proves to be very useful for people with

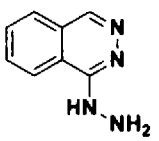
renal dysfunction. Its exact mechanism remains unknown. Hydralazine is a stable yellow solid with low water solubility (~3%). It reaches peak plasma concentration within 1 hour and is rapidly metabolized by hydroxylation, glucuronidation, and *N*-acetylation.

Sodium nitroferricyanide (or sodium nitroprusside, Figure 26) is one of the most potent blood pressure–lowering drugs, but its use is restricted to emergencies because of its short duration of action. Dilation of arterial and venous SMCs occurs. The exact mechanism remains unknown, but most probably it decomposes and releases NO, which results in nonspecific dilation of blood vessels.



Mannitol

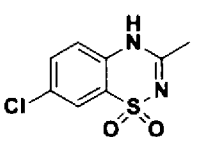
Figure 25. *A class V osmotic diuretic, mannitol.*



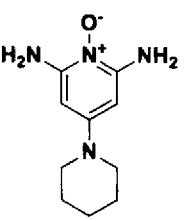
Hydralazine

$Na_2[Fe(CN)_5NO]\cdot 2H_2O$

Sodium Nitroprusside



Diazoxide



Minoxidil

Figure 26. *The structures of several direct-acting vasodilatory agents.*

Diazoxide and minoxidil (Figure 26) are potassium channel agonists that decrease the concentration of calcium ions in the cells and, thus, reduce the excitability of the SMCs. Both lower the peripheral vascular resistance; however, minoxidil requires activation by sulfotransferase to minoxidil sulfate before it becomes functionally active. Diazoxide must be injected intravenously (pH 11.5), which converts the drug to its soluble form. The drug is highly protein bound and can displace other drugs. Minoxidil is useful in severe hypertension that is difficult to control with other antihypertensive drugs.

Anticoagulation

The blood circulatory system has to be self-sealing, otherwise continued blood loss from even a small injury would be life threatening. Under normal circumstances, bleeding is rapidly stopped by hemostasis. Technically, hemostasis is a combination of many processes arising from physical and chemical interactions between soluble components of the plasma, the vascular bed, and cellular material. The final result of these interactions is the formation of a thrombus, a highly cross-linked insoluble mass containing cells, enzymes, and other proteins at the site of injury that stops blood loss and ingress of microbes into the vasculature. Intravascular thrombosis can be caused–without external injury–by vascular injury and by blood hypercoagulability.

Physical and Chemical Forces Involved in Clotting

On injury to the vascular wall, the subendothelial cells are exposed, setting off a series of responses. First, a reduction in blood flow occurs through vasoconstriction, which allows platelet adhesion to the cells at the site of injury (**Figure 27**). Simultaneously, certain chemicals that facilitate platelet activation and aggregation are released. Platelet aggregation leads to the formation of a plug in which a mass of platelets are held together by physical noncovalent forces. The platelet plug is susceptible to rupture by shearing forces due to rapid blood flow. Stabilizing this platelet plug is the formation of fibrin, a 3-dimensional, covalent, polypeptide network, which is the end result of chemical processes forming the coagulation cascade. Molecules that do not allow platelets to aggregate and, thus, prevent clotting, especially in the arteries, are called antiplatelet agents. These include aspirin, dipyridamole, and ticlopidine. Molecules that disintegrate a preformed clot constitute *fibrinolytic agents.* A typical example in this category is the enzyme, streptokinase.

The chemical forces involved in clotting include a series of chemical reactions mediated by proteins, enzymes, calcium ions, and phospholipid surfaces. The series of steps mimics a cascade of events that is traditionally described as a bifurcated process consisting of intrinsic and extrinsic pathways. The intrinsic pathway uses factors soluble in the plasma only, whereas the extrinsic pathway consists of some factors that are insoluble in the plasma, eg, membrane-bound factors (factor VII).

The distinguishing feature of the coagulation cascade is that activation results in an enzyme that converts the proenzyme (zymogen form) in the next step to an active enzyme. The newly formed enzyme then acts further down the cascade to greatly amplify the initial activation signal. These reactions occur in the presence of Ca^{+2} ions and on an appropriate phospholipid membrane.

The intrinsic and extrinsic coagulation pathways merge at the formation of factor Xa (Figure 27), which, in turn, cleaves prothrombin at 2 sites to yield the key enzyme of the clotting cascade, thrombin. Thrombin has numerous properties, chief among them are coagulation and anticoagulation activities. Under appropriate conditions, thrombin cleaves fibrinogen, which constitutes nearly 2% to 3% of plasma protein. Fibrinogen cleavage results in soluble fibrin monomers that spontaneously aggregate to form a soft clot. This aggregate is rapidly converted to a more stable "hard clot" by the covalent cross-linking of neighboring fibrin molecules in a reaction catalyzed by factor XIIIa (fibrin-stabilizing factor). The process of 3-dimensional cross-linking that occurs in the polymeric fibrin formation traps numerous cells, including red blood cells and platelets.

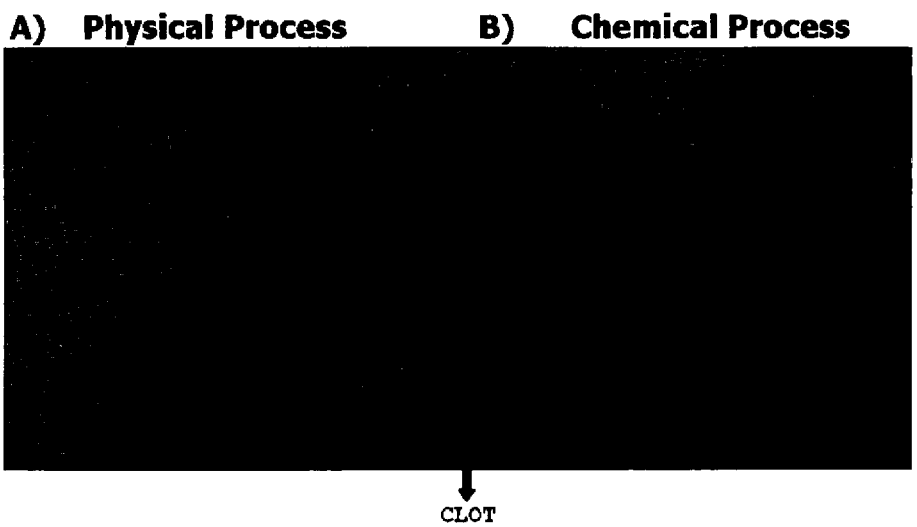


Figure 27. *Key elements of clotting consisting of physical and chemical processes. Both processes are more complex than depicted here and are not completely independent, as shown here. There is significant cross-talk between the two, exemplified by thrombin that is involved in the activation of platelets. Inhibitors of clotting are shown in boxed text with arrows indicating their site of action. From Desai UR. New antithrombin-based anticoagulants. Med Res Rev. 2004;24(2):151-181 with permission. Abbreviations: ADP, adenosine diphosphate; GP, glycoprotein; Glu, glutamic acid.*

Anticoagulants

Anticoagulants are molecules that prevent blood clotting by inhibiting the chemical process of proteolytic formation of the 3-dimensional fibrin polymer. These agents include the heparins (heparin, low-molecular-weight [LMW] heparin, fondaparinux), warfarin, the hirudins (desirudin and bivalirudin) and the peptidomimetics (argatroban and dabigatran).

Anticoagulants are indicated in myocardial infarction, venous thrombosis, peripheral arterial embolism, pulmonary embolism, and many other conditions. Anticoagulants are also used in blood transfusions, extracorporeal blood circulation, and dialysis procedures. The following discussion pertains to the structural, functional, and physicochemical aspects of clinically used anticoagulant drugs.

The Heparins

Heparin is a strongly acidic, high-molecular-weight mucopolysaccharide with rapid anticoagulant effects. Heparin, prepared from porcine lung or intestinal mucosa, is a mixture of molecules of varying molecular weights and chemical structures. It is a linear polysaccharide composed of alternating residues of glucosamine and uronic acid that are linked in a 1→4 manner (**Figure 28**). A typical preparation of heparin may have polysaccharide chains in the range of 10 to 100 monosaccharide residues corresponding to an average molecular weight (M_R) range of 3,000 to 30,000. The uronic acid residues may be of the β-D-glucuronic or α-L-iduronic type. The β-D-glucosamine residues are typically sulfated ($-OSO_3^-$) or acetylated ($-COCH_3$) at the 2 position. The available hydroxy groups, such as the 2-OH in iduronic acid and the 6-OH in β-D-glucosamine, may be sulfated. The numerous negatively charged groups ($-COO^-$ and $-OSO_3^-$) that span the length of the linear polysaccharide give a strongly acidic character to this molecule. In fact, heparin is the strongest acid in the body. Thus, heparin is a drug with massive structural microheterogeneity and polydispersity. The polyanionic character of heparin is probably the single major source of its numerous adverse effects.

Low-molecular-weight heparins (Figure 28) obviate some of these adverse effects. Low-molecular-weight heparins, prepared by chemical or enzymatic depolymerization or by chromatographic resolution of full-length heparin have an M_R range of 3,000 to 8,000 with an average of 5,000 corresponding to about 15 monosaccharide residues. Reduction in the M_R substantially reduces their negative charge density, thereby lowering their nonspecific interactions with plasma proteins and cells. However, the chemical and enzymatic methods used to prepare LMW heparins may introduce additional structural variations to increase the heterogeneity of the commercial preparations.

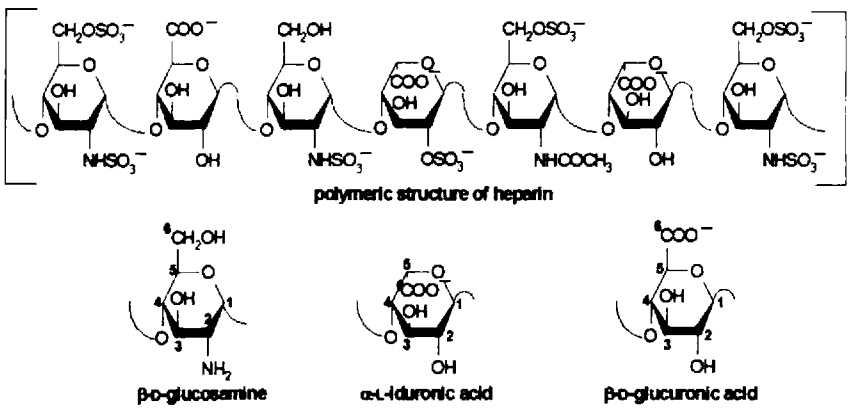


Figure 28. *Structure of the polymeric heparin sequence and its constituent monosaccharide residues. Unfractionated heparin has an average molecular weight of 15,000 Da, whereas low-molecular-weight heparin is about 3 to 5 times smaller.*

The key structural unit of heparin is a unique pentasaccharide sequence (**Figure 29**). This sequence is also the basis of fondaparinux, a synthetic drug introduced into clinical use in 2001. The generic 5-residue heparin pentasaccharide sequence consists of 3 D-glucosamine and 2 uronic acid residues. The central D-glucosamine residue contains a unique 3-*O*-sulfate moiety that is rare outside of this sequence. Four sulfate groups on the D-glucosamines (Figure 29) are critical for retaining high anticoagulant activity. Elimination of any one of them results in a dramatic reduction in the anticoagulant activity. Removal of the unique 3-*O*-sulfate group results in complete loss of the anticoagulant activity. Removal of sulfate groups other than the critical ones seems to insignificantly affect anticoagulant activity. Only a third of the chains in commercial heparin preparations have this unique pentasaccharide sequence. Thus, more than two-thirds of heparin chains are probably not active as anticoagulants. The LMW heparin preparations may have considerably varying proportions of chains with the active site.

Properties of Heparin

Because of its highly acidic sulfate groups, heparin (or LMW heparins) exists as a polyanion at physiological pH. The heparin polysaccharide chain is degraded by gastric acid and must, therefore, be administered intravenously or subcutaneously. Because of its smaller size, LMW heparin is more bioavailable when given subcutaneously. Heparin is typically not given intramuscularly because of the danger of hematoma formation. Peak activity of heparin is reached within minutes of administration and is found to last 2 to 6 hours after intravenous administration and 8 to 12 hours after subcutaneous administration.

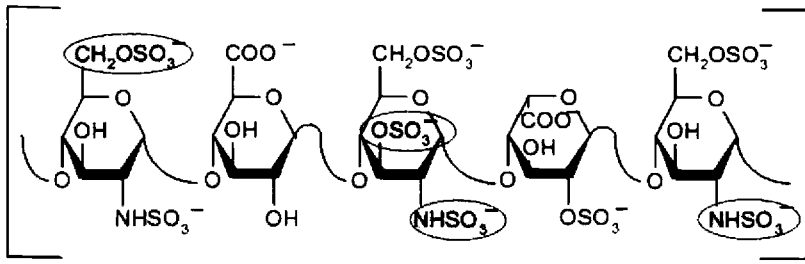


Figure 29. *Structure of the pentasaccharide sequence of heparin that binds to antithrombin. Negatively charged groups highlighted in ovals are critical for high-affinity binding and activation of antithrombin for accelerated inhibition of factor Xa and thrombin. Synthetic fondaparinux is derived from this high-affinity heparin pentasaccharide sequence.*

Fondaparinux has a much longer half-life (~16 h) because it remains bound to the antithrombin present in plasma. Heparin is relatively nontoxic and can be safely used in pregnancy because it does not cross the placental barrier. Heparin overdose or hypersensitivity may result in excessive bleeding. If hemorrhage occurs, the anticoagulant effect of heparin can be reversed in minutes by administration of protamine sulfate, an LMW protein that has multiple positively charged groups. In contrast, fondaparinux activity cannot be reversed with protamine sulfate. In fact, no effective antidote for fondaparinux is available, which generates substantial risk in case of accidental bleeding. In addition to bleeding complications, unfractionated heparin has substantial intrapatient and interpatient dose variability, which requires frequent laboratory monitoring. The inconsistent response is due to differences in the bioavailability and from heparin binding to proteins in the plasma and on cells. Some of these heparin-binding proteins are acute phase proteins, which are typically elevated in very ill patients, thus reducing heparin efficacy. In addition, heparins do not neutralize clot-bound thrombin, thus raising the possibility of reactivating clotting at a later time. Finally, heparin-induced thrombocytopenia (HIT) is a potentially lethal complication of heparin therapy that may be associated with thrombosis. Heparin-induced thrombocytopenia involves a substantial drop in the platelet count after the initiation of therapy.

Biochemical Mechanism of Heparin Action

The anticoagulant action of heparin occurs through *antithrombin,* a glycoprotein that exists in plasma at a fairly high concentration (~2 μmol/L). Antithrombin is an inhibitor of many proteases of the coagulation cascade, especially thrombin and factor Xa, under physiological conditions. Thus, antithrombin prevents the conversion of fibrinogen to fibrin, thereby inhibiting clotting. However, the rate of inhibition of both of these enzymes by antithrombin is rather slow

(**Figure 30**). Heparin, containing the unique 5-residue sequence, forms a high-affinity complex with antithrombin. The formation of the antithrombin-heparin complex increases the rate of inhibition of factor Xa and thrombin some 500- to 4,000-fold (Figure 30). Heparin chains devoid of the pentasaccharide sequence cannot bind antithrombin tightly and, hence, do not increase the inhibition of the critical coagulation enzymes. The fact that the heparin-antithrombin complex inactivates proteolytically active molecules, thrombin and factor Xa, suggests that heparin action is instantaneous. Accelerated inhibition of factor Xa and thrombin shuts down the coagulation process and is the primary reason for the clinical use of heparin as an anticoagulant.

The Coumarins and Indanediones

The most common coumarin is warfarin (**Figure 31**), which was discovered in the 1930s while trying to identify a coagulant. Coumarin and 1,3-indanedione are small, aromatic, water-insoluble scaffolds. While coumarin contains a lactone moiety, 1,3-indanedione contains 2 ketone groups. Introduction of a 4-OH substitution in the coumarin structure and a 2-aryl substitution in the 1,3-indanedione structure improves the water solubility of these scaffolds because of resonance stabilization of the anion formed under alkaline conditions (**Figure 32**).

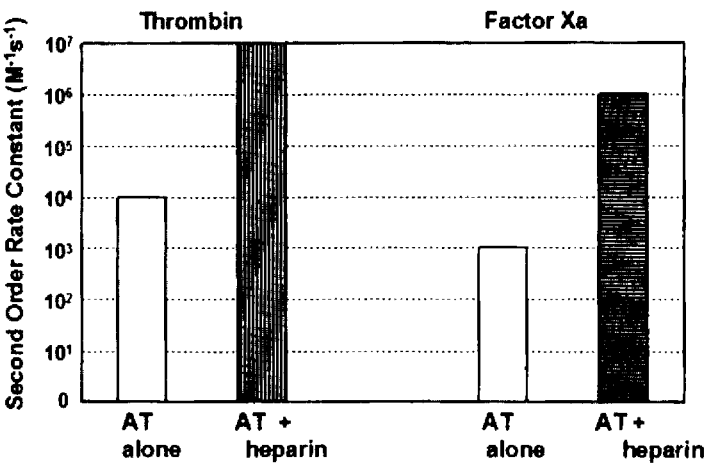
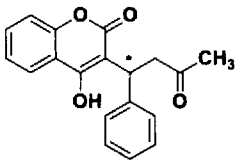


Figure 30. *Heparin accelerates the rate of antithrombin (AT) inhibition of procoagulant enzymes. The second-order rate constant increases from approximately 10^3 and 10^4 for factor Xa and thrombin, respectively, with antithrombin alone to ~10^6 and 10^7 in the presence of heparin. Note the logarithmic scale on the y-axis. See text for details.*



Warfarin
* Chiral center

Figure 31. *Structure of warfarin.*

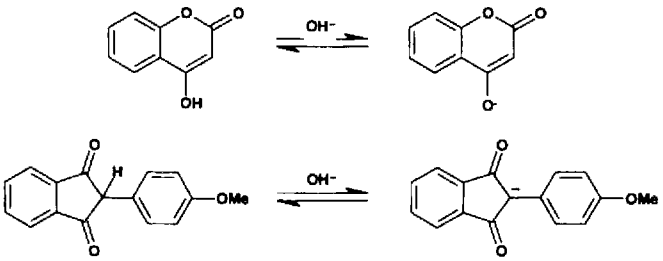


Figure 32. *Weakly acidic property of 4-hydroxycoumarins and 2-aryl-1,3-indandiones.*

Neither coumarin nor 1,3-indanedione have anticoagulant activities. However, the 4-OH and 3-alkyl substitutions in the coumarin ring and the 2-aryl substitution in the 1,3-indanedione ring confer anticoagulant activity. The coumarin derivatives are used to treat thrombophlebitis, pulmonary embolism, and coronary thrombosis. Some coumarins have been used as rodenticides.

The coumarins are the most often used oral anticoagulants. Another characteristic of the coumarins is that they have a very slow onset of action. They typically exert their effects in vivo only after about 24 to 48 hours, and their duration of action is also much longer, about 1.5 to 5 days. The reason for this difference in activity lies in their biochemical mechanism of action.

Biochemical Mechanism of Action of Coumarins

Coumarins and 1,3-indanediones are competitive inhibitors of vitamin K in the biosynthesis of prothrombin. They prevent the formation of prothrombin that contains 10 γ-carboxyglutamic acid residues that are crucial for the optimal activity of thrombin formed by the action of factor Xa.

For thrombin to be fully functional as a clotting enzyme in the presence of Ca^{+2} and phospholipids, it should have 10 γ-carboxyglutamic acid (Gla) residues in its *N*-terminal end. These Gla residues bind several Ca^{+2} ions under physiological conditions, resulting in a specific conformation of the molecule that is important for fibrinogen recognition and cleavage activity.

Under normal circumstances, 10 glutamic acid (Glu) residues in prothrombin are γ-carboxylated to give a prothrombin molecule in a posttranslational modification reaction that is catalyzed by vitamin K (**Figure 33**). Thus, vitamin K is required in the hepatic biosynthesis of prothrombin that contains 10 γ-carboxyglutamic residues. Only this prothrombin molecule containing 10 Gla residues on activation with factor Xa results in a thrombin molecule that is fully functional under physiological conditions.

The 3-dimensional structure of anticoagulant coumarins and 1,3-indanediones resembles vitamin K; thus, they compete for the vitamin K binding site on vitamin K epoxide reductase and vitamin K reductase, resulting in inhibition of γ-carboxylation of the Glu residues in prothrombin. Thus, an abnormal prothrombin is synthesized in the presence of these anticoagulants that, when activated to thrombin, has less than 2% of the proteolytic activity for fibrinogen as normal thrombin. This inhibition of the formation of Gla residues is not restricted to prothrombin alone and is likely to occur for all zymogen forms of the enzymes of the coagulation cascade, thus enhancing the potency of the oral anticoagulants.

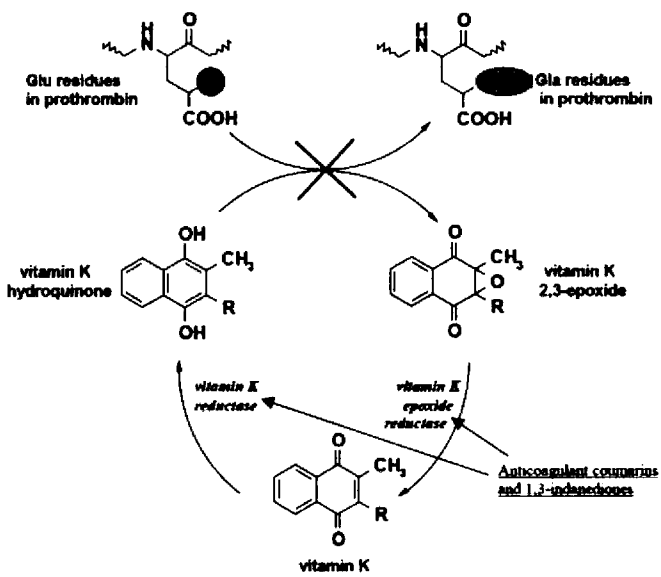


Figure 33. *Importance of vitamin K in the γ-carboxylation of glutamic acid (Glu) residues of prothrombin and inactivation of the process by anticoagulant coumarins and 1,3-indandiones. These anticoagulants inhibit vitamin K reductase and vitamin K epoxide reductase, enzymes with an important role in the oxidation-reduction process necessary for γ-carboxylation of prothrombin. Abbreviation: Gla, γ-carboxyglutamic acid.*

Coumarins and 1,3-indanediones do not inactivate thrombin and factor Xa that might have been formed before the administration of the oral anticoagulants. Thus, the antithrombotic effect of these molecules is not observed immediately. The observed slow onset is due to the time required for the body to eliminate predrug prothrombin blood levels (half-life, ~2.5 days). The longer duration of action is because of the time required by liver to resynthesize normal prothrombin following the suspension of oral anticoagulation therapy. Coumarins and 1,3-indandiones interact with certain drugs. For example, the action of oral anticoagulants can be enhanced by phenylbutazone and salicylates and antagonized by barbiturates and vitamin K. Another major risk factor for warfarin therapy is interaction with constituents of food. Spinach and collard greens (or green leafy vegetables) that contain substantial quantities of vitamin K affect the anticoagulant potency of warfarin when consumed in large quantities. Likewise, warfarin taken with cranberry juice, which contains several flavonoids known to bind many CYP isoenzymes, can have metabolic adverse effects.

The Hirudins

While heparins and coumarins are indirect inhibitors of thrombin, the hirudins are direct thrombin inhibitors (DTIs). *Hirudin,* isolated in 1884 from the leech *Hirudo medicinalis,* is a polypeptide with 65 amino acid residues. Hirudin is the most potent thrombin ligand known to date, with a K_I (inhibition constant) value of about 20 fM. Hirudin binds in an allosteric site of thrombin called anion-binding exosite 1 and in the active site of thrombin, thereby preventing thrombin hydrolysis of fibrinogen, which also has to bind in both of these sites for conversion to fibrin monomers. This process leads to the inhibition of clot formation. Yet, the very high affinity of hirudin for thrombin is problematic because of frequent bleeding episodes in patients during hirudin therapy. To reduce these problems, lepirudin and desirudin were designed, in which 2 residues at the *N*-terminus of hirudin are replaced with leucine-threonine and valine-valine, respectively, and tyrosine 63 does not bear the sulfate group present in native hirudin. Even these 2 hirudin analogs are associated with bleeding issues. Thus, a newer and smaller class of hirulogs, analogs of hirudin, called bivalirudin, was developed. It consists of 4 *N*-terminal residues and 12 *C*-terminal residues of hirudin connected by 4 glycine residues.

The major bleeding issues of the hirudins limit their use to patients with disease refractory to unfractionated heparin and patients with HIT. Another common problem is that they have a narrow therapeutic window. Hirudins also have some immunogenic risk because they are foreign peptides. The very high affinity of hirudin, lepirudin, and desirudin for thrombin makes them essentially irreversible drugs. There is no effective antidote available for reversing these complexes. Bivalirudin (**Figure 34**) binds thrombin with a K_I of 1 to 2 nM, which is much weaker than hirudin. Yet, bivalirudin has a short half-life (25 minutes) because

thrombin cleaves its arginine 3–proline 4 bond. Bivalirudin is considered superior to other hirudins because of its more predictable anticoagulant response and its lower immunogenicity. It is also considered a safer anticoagulant for hepatically and renally compromised patients because it can be hemodialyzed.

The Peptidomimetics

Peptidomimetics are small, organic molecules that mimic the key peptide scaffold of a polypeptide therapeutic agent. Argatroban (**Figure 35**) is the first rationally designed, small-molecule, peptidomimetic inhibitor to reach the clinical setting; dabigatran etexilate is currently the only DTI available for oral use. It is a double prodrug form of dabigatran, which is a benzimidazole-based small-molecule inhibitor. Argatroban, approved for use in patients at high risk for HIT, is limited by its short plasma half-life (24 minutes). The 2 DTIs specifically target the active site of thrombin and, thus, function as competitive inhibitors. Thrombin recognizes an arginine at the P1 position, while prefers hydrophobic amino acids such as proline and phenylalanine at the P2 and P3 positions, respectively. (*Note:* The amino acid residue on the N-terminal side of the scissile bond is identified as P1 residues. Residues following the P1 residue on the N-terminal side of the scissile bond are called P2, P3, and so on.) The peptidomimetics typically contain a guanidine (or an amidine) moiety to mimic the P1 arginine side chain and an organic scaffold such as an alicyclic, aromatic, or heteroaromatic ring connected through amide or sulfonamide linker(s) to mimic the proline and phenylalanine residues at the P2 and P3 positions.

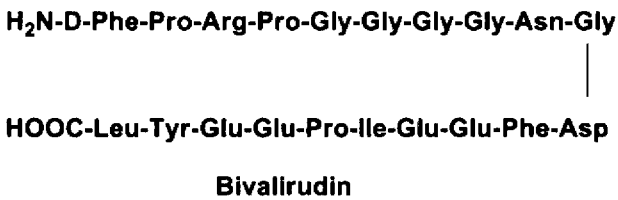
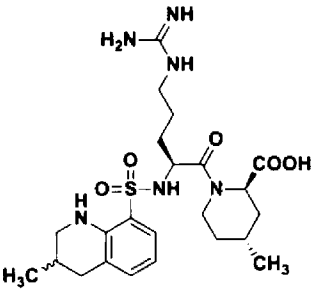
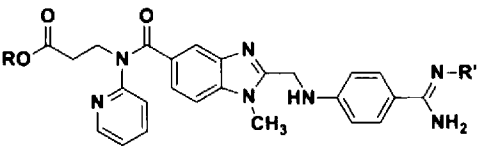


Figure 34. *The sequence of bivalirudin. Abbreviations: Asp, aspartic acid; Glu, glutamic acid; Ile, isoleucine; Leu, leucine; Phe, phenylalanine; Pro, proline; Tyr, tyrosine.*



Argatroban



Dabigatran etexilate R = $-CH_2CH_3$; R' = $-COO-(CH_2)_5-CH_3$
Dabigatran R= -H; R' = -H

Figure 35. *Selected peptidomimetics that act as thrombin inhibitors.*

Suggested Reading

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Discussion Topics

1. Describe the targets and effects of the following cardiovascular drugs:

   a. antianginal drugs

   b. antiarrhythmic drugs

   c. antihypertensive drugs

   d. anticoagulant drugs

2. Several agents prescribed as cardiovascular agents have multiple modes of action. Discuss agents falling into this category and which effects predominate.

3. As a healthcare provider, you must be aware of adverse effects your patients may have as a result of taking common medications. Discuss what these effects are and how you would moderate those effects.

4. Discuss what effects certain foods might have when combined with therapy, eg, anticoagulant therapy with warfarin.

5. In 2008, a major issue arose in relation to the contamination of clinically used heparin with oversulfated chondroitin sulfate. Discuss the possible structural reasons for contamination of heparin not being detected early enough.

6. The widely used drug sildenafil is an inhibitor of phosphodiesterase, which preserves the levels of nitric oxide in the bloodstream, resulting in muscle relaxation. Why is sildenafil not used as an antianginal agent?